Clovis SFEIR

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Profile

IMT Atlantique engineering student combining numerical simulation and AI automation. Built a finite-difference PDE solver (OpenMP, **2nd-order convergence** verified against a closed-form solution) and, around coding agents, a Rust autonomous delivery harness at Infotel's AI Lab (928 tests) and a supervisor that runs an agent unattended under budgets, watchdogs and independent review.

Education

IMT Atlantique – Engineering Degree in Software Engineering (Apprenticeship) *Sep 2025 – Jun 2028*
Nantes, France *Algorithms & Data Structures, OS, Computer Networks, Distributed Systems, Cloud Computing, AI*

Bachelor in Computer Science *Sep 2022 – Jun 2025*
La Rochelle, France ***Class rank: 1/85*** *| Software Engineering, Databases, Cybersecurity*

HRT Macro Placement Challenge 2026 (Hudson River Trading/Partcl) — **24th/124** (top 20%); hybrid analytical placer: force-directed global placement with timing-aware legalization; proxy cost **1.095** beats SA baseline (2.125) and RePlAce (1.458); field includes Stanford, Princeton, ETH Zürich.

Projects

Reservoir Diffusivity PDE Solver *C++20, OpenMP, finite-difference*

- Explicit FTCS and implicit Crank-Nicolson finite-difference schemes for the reservoir diffusivity (pressure-transient) equation; verified via grid-refinement convergence study against the closed-form erfc analytical solution – error ratio converges to **4.0** as dx halves (textbook 2nd-order accuracy) across 5 refinements.
- OpenMP-parallelized stencil/assembly loops; measured (not assumed) scaling identifies the Amdahl's-law-bounded sequential Thomas solve and memory-bandwidth-bound behavior below L3 size; 13 test cases, 40,300 assertions; Docker image runs the full suite at build time.

Qwen Autonomous Supervisor *Python, agent reliability, long-running automation*

- Durable supervisor that keeps a coding agent running unattended in bounded ticks: wall-time/tool-call/token/cost budgets, exclusivity leases, watchdogs that kill runaway process trees, clean-worktree checks around mutations, and a mandatory independent review session.
- Resumable endurance campaigns (4–12 h) with deliberate agent-crash injection; campaign state is checkpointed in a durable ledger so a killed supervisor resumes the same run.

CCUS Injection MRV *Python, Peng-Robinson EOS, Monte Carlo, CUSUM*

- CO_2 density from a Peng-Robinson equation of state with a volume-shift calibrated on one CoolProp point: error $\leq$1.4% away from the critical point, $\approx$6% near it (measured and disclosed); cumulative injected mass with Monte Carlo uncertainty (1,000 draws, $\pm$2.1%).
- CUSUM leak detector on an injectivity model: precision **0.993**, recall **0.965**, detection delay 0.83 h on a labeled synthetic leak; FastAPI service, 53 tests, Docker build verified in CI.

Renewable Forecast + Dispatch Optimizer *Python, gradient boosting, linear programming*

- Next-hour wind forecast (gradient boosting, strictly time-ordered split, lag-feature leakage tests) feeding a linear-programming economic dispatch (HiGHS) with power-balance, generator-limit and ramp-rate constraints.
- Forecaster beats the 24 h seasonal-persistence baseline on RMSE (**7.54 vs 7.97 MW**) with MAE on par (5.36 vs 5.33); quantifies the cost of forecast error at $30,073 vs perfect foresight; 17 tests.

Work Experience

Infotel *Nantes, France*
AI Engineer Apprentice – Lab IA *Sep 2025 – Jun 2028*

- Built a Rust-based autonomous software-delivery harness that takes a complete specification through architecture synthesis (subsystems, interface contracts), epic/slice decomposition, and TDD-first generation to a tested, deployable application end-to-end; automated compile+test quality gates with iterative self-repair, independent code-review pass, and cross-slice conflict detection before delivery; 928 Rust tests; validated end-to-end producing a fully working, 100%-test-passing application directly from a single specification.

Excelia Group *La Rochelle, France*
Software Developer Intern / AI Software Engineer Intern *Two internships: Jan–Mar 2025 & May–Jul 2024*

- RAG pipeline (chunking, OpenAI embeddings, flat FAISS, cross-encoder re-ranking): **40% reduction** in lookup time vs. keyword-search; cut p95 latency 30–40% via embedding cache + asyncio parallelisation; Prometheus/Grafana, Docker, GCP.
- Real-time interview simulator (Next.js/React, GPT-4o, ElevenLabs TTS): streaming responses, scenario-branching state machine, A/B prompt-scoring harness.

Technical Skills

Perf / Systems	OpenMP, cache-aware layout, acquire/release atomics, ICS/SCADA protocol parsing (Modbus, IEC-104)
Languages	C++20, Python, Rust, TypeScript, SQL
Numerics / AI	finite-difference PDE, Monte Carlo, Peng-Robinson EOS, linear programming (HiGHS), PyTorch/LoRA, agent automation, BM25
Infrastructure	Docker, tokio async, Prometheus/Grafana, GitHub Actions CI, GCP, Redis, ZeroMQ
Frameworks	FastAPI, Flask, Node.js/Express, React/Next.js, REST/gRPC